ADVANCED EMBEDDED SOFTWARE DESIGN

System Requirement Specification for Mars Rover Application

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1. Introduction

Our elective project requires the group to create software to control a robot on Mars, which must then explore Mars. The goal of this project is to create a robot that can operate autonomously in challenging circumstances. Our goal is to launch this robot to Mars, where the climate is inhospitable to human survival. It will gather information from the planet and transmit it to researchers on Earth so they can learn more about it. The requirements that were compiled as a group are presented below, along with a summary of each component.

1.1 Purpose

System requirement specifications for the Mars Rover are meant to provide a framework for the team to work within. It is written for the team to create it in accordance with the specification so that the robot may be built with the proper functionality. Here, the reasoning underlying the inputs and outputs is briefly discussed. The need specification further elaborates the specifications, factors, elements, characteristics, and behavior of the robot that are essential for guaranteeing the precise performance of the rover.

1.2 Scope

The accuracy of the Mars Rover must be maintained in all difficult and obstructed circumstances. The rover features an engine to move it from place to place and a camera to record its surroundings. The operator sends certain Telecommand (TC) blocks to the rover, which then carries them out as directed. The rover returns the output as a Telemetry (TM). Only two zones exist: the safe zone and the danger zone. When the rover is in a safe zone, it may move around and use its camera to record the environment. When it is in a danger zone, however, it can still move around but the camera's lid is closed for the camera’s protection. In addition, the rover's condition is continuously updated for its precise operation.

1.3 List of Abbreviation

Table 1:List of abbreviations

Abbreviation	Explanation
HHSLP	HAW-Hamburg Safe Layer Protocol

1.4 References

HHSLP

2. System Architecture

The general architecture of Mars Rover application is shown in Figure 1. The Rover's onboard computer receive command from the operator so that it can perform some actions like open lid, engine on, change direction etc. The rover can also communicate with the operator. The system has x and y sensors to locate the position of rover. Safe communication between the O/C and Operator and other components is assured by the HAW Hamburg Safe Layer Protocol.

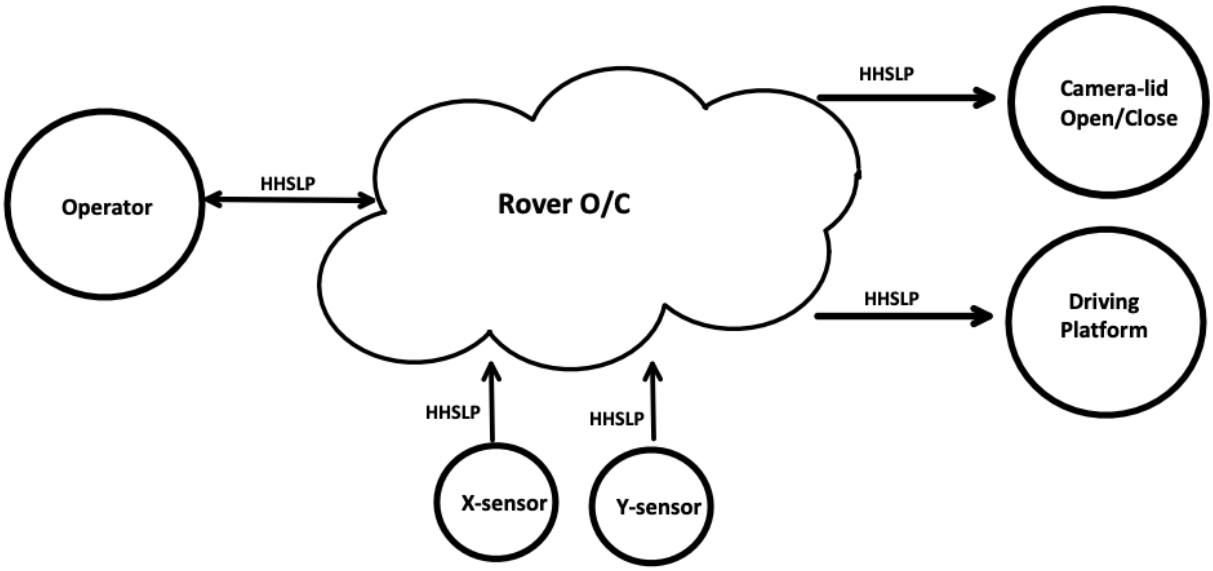

Figure 1:Architecture of Mars Rover application.

The Mars Rovers Interface description can be seen in Figure 2.

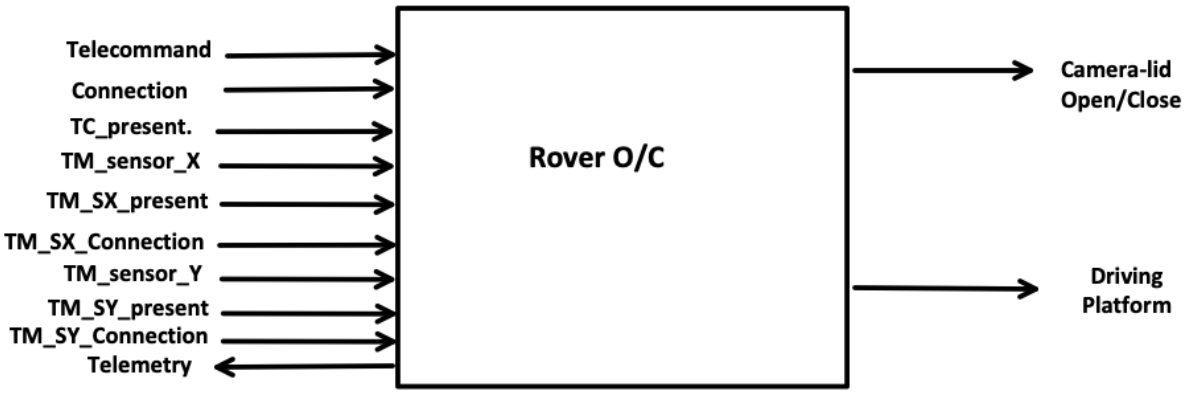

Figure 2: Block diagram of Mars Rover.

3. Requirements

3.1 General Operations

#001. **Mars Rover Input:**

The rover has the following inputs-

1. Connection.
2. TC_present.
3. TC.
4. TM_sensor_X.
5. TM_SX_present.
6. TM_SX_Connection.
7. TM_sensor_Y.
8. TM_SY_present.
9. TM_SY_Connection.

#002. **Mars rover Output**:

The rover has the following output-

1. Camera lid.
2. Driving Platform.
3. Telemetry.

#003. "*Engine on*" Command

System requires input from operator to which target it should move.

Conditions must be given:

⇒ Connection to operator must be established

⇒ Operator command must be present

System functionality:

System must store information about

⇒ That it must move (move-variable set to true)

⇒ Which target it should move to (target-variable set to [x, y])

#004. "*Engine off*" command

Conditions must be given:

⇒ Connection to operator must be established.
⇒ Operator command must be present.

System functionality:
⇒ System must set move-variable to false

#005. "*Close-lid*" command

Conditions must be given:
⇒ Connection to operator must be established.
⇒ Operator command must be present.

System functionality:
⇒ System must set lid_open-variable to false.

#006. "*Open-lid*" command

Conditions must be given:
⇒ Connection to operator must be established.
⇒ Operator command must be present.

System functionality:
⇒ System must set lid_open-variable to true.

#007. "*Set-danger-zone*" command

System requires input from operator where the center of the zone is located and which radius the zone has.
Conditions must be given:
⇒ Connection to operator must be established.
⇒ Operator command must be present.

System functionality:
⇒ System must set danger_zone_centre-variable (as array with coordinates).
⇒ System must set danger_zone_radius-variable.

#008. "*Set-x-position*" command

System requires input from x-sensor where the rover is located.
Conditions must be given:
⇒ Connection to x-sensor must be established.
⇒ announcement command must be present.

System functionality:

⇒ System must set x_position-variable.

#009. "*Set-y-position*" command

System requires input from y-sensor where the rover is located.
Conditions must be given:

⇒ Connection to y-sensor must be established.

⇒ announcement command must be present.

System functionality:

⇒ System must set y_position-variable.

#010.

Rover's Communication:
The Mars Rover should receive command from the Operator only.

#011.

Station's Communication:
The Mars Rover should be able to send and receive messages

#012.

The Rover should cyclically be able to send its status to the operator.

#013.

The Rover should be able to inform the operator about the camera lid status, engine status etc.

#014.

The Rover should be able to continue sent its current to the operator when no new valid messages is received.

3.2 Requirements for the Camera lid

#015. In the initial state the O/C sends a TC to the camera each time the connection to it is established.

#016. The lid should be closed in the danger zone.

#017. The state of camera lid is "*open*" if and only if the following conditions are met:

⇒ The rover is outside the danger zone.

⇒ The operator has sent a command to enable the opening of the camera lid.

Otherwise, the lid should be "*closed*".

#018. The state of the lid is computed in every cycle.

3.3 Requirement for the Danger Zone

#019. In the initial state, The operator sends a command for the danger zone setup. The danger zone can be anywhere.

#020. For the danger zone, The operator sets x and y position and through the x and y sensors. In case any one of the sensors fail-

⇒ When x-sensor fails, the rover identifies the danger zone a rectangular region which is parallel to y-axis and the rover could also be in the danger zone.

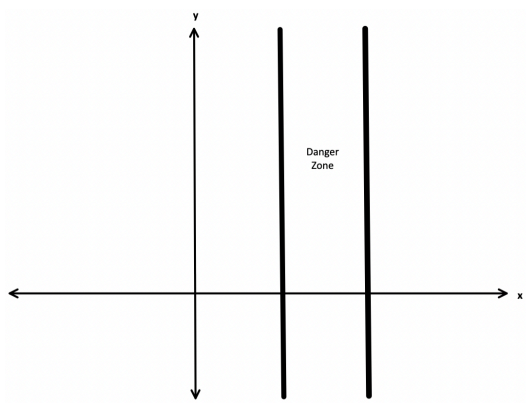

Figure 3: Danger zone when x-sensor fails.

⇒ Similarly, when y-sensor fails, the rover identifies the zone a rectangular region that is parallel to x-axis and the rover could also be in the danger zone.

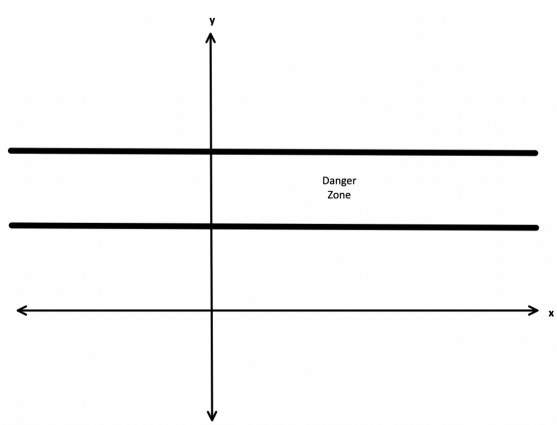

Figure 4:Danger zone when y-sensor fails

#021. The position of rover is identified by the x and y sensors.

#022. The position of the rover cannot be traced and the engine is turned off if any of the following events occur:

⇒ If x-sensors fails.

⇒ Or If y-sensors fails.

⇒ Or If both sensors fail.

#023. Inside the danger zone, the engine is on and the rover can move.

#024. The camera lid is closed inside the zone.

3.4 Requirement for the Engine

#025. In the initial state, The o/c should send the TC to the driving platform each time the connection to it is established. The command may be "*start*" or "*stop*" to start or stop the engine respectively.

#026. In any other state, the O/C sends command to the engine only if the state of the engine is to be change.

#027. After power-up the O/C does not send any movement command to the engine.

#028. When the operator sends the TC about the target position, the O/C sends the movement command to the engine.

#029. The state of the engine does not depend on the danger zone. Rather it depends on the operator's command.

3.5 Mars Rover and HHSLP Connection Lost

#030. **Connection Lost:**

The state of rover is unknown if

⇒ if the O/C and HHSLP link were broken.

⇒ Also, if O/C signals "undefined" for its own state

#031. **HHSLP Reconnection:**
When the HHSLP connection is lost, the O/C should try to re-establish it.

4. Mars Rover Interface Requirements

The Mars rover must be able to receive packets containing TC from the operator. The O/C then unpacks the packet, decrypts the packet containing the TC, and returns the response to the operator.

#000. **Package delimiter:**

Each received and sent packages should start and end with the hexadecimal value “0x7E” as a delimiter between two serial packages.

“0x7E”	Package Content	“0x7E”

4.1 Input Message Format

Input messages are received by O/C from the operator. These messages are received as packages. Then they are correctly interpreted and divided into simple messages. Then these messages are used as inputs to the O/C.

Package Content

“0x7E”	Lid Open enable	Danger Zone Set-up	Engine on/off State	“0x7E”

Figure 5: Input message Format.

#001. **Input Package Size:** The size of the input package is 5 Bytes.

#002. **Input Package Data Types:** The data type of input message is uint16 array of length 3.

#003. **Input Message Length:** The length of each input message inside the package is 1Byte.

4.2 Output Message Format

The O/C should communicate with the operator about the camera lid status, driving platform and danger zone set-up. All the information are sent as a single package using a suitable data type. An example is given below_

	Lid status “open”	Danger zone not set	Engine “off”	
“0x7E”	0x01	Null	0x00	“0x7E”

Figure 6: Output message format.

#004. **Output Package Size:** The size of the output package is 5 Bytes.

#005. **Output Package Data Types:** The data type used here is uint16 of array 3.

#006. **Output Message Length:** The length of output message is 3 Bytes.

4.3 Command Codes

<table>
<tr><th>Command</th><th>Command Code</th><th>Arguments</th></tr>
<tr><td>Lid Open enable</td><td>0x1</td><td><u>1 Byte:</u>

0x0: Disable.
0x1: Enable.</td></tr>
<tr><td>Danger Zone Set-up</td><td>0x2</td><td><u>13 Bytes:</u>

<table><tr><td>Byte 0</td><td>enable/disable.</td></tr><tr><td>Byte 1</td><td rowspan="4">Set radius</td></tr><tr><td>Byte 2</td></tr><tr><td>Byte 3</td></tr><tr><td>Byte 4</td></tr><tr><td>Byte 5</td><td rowspan="4">Set x-position</td></tr><tr><td>Byte 6</td></tr><tr><td>Byte 7</td></tr><tr><td>Byte 8</td></tr><tr><td>Byte 9</td><td rowspan="4">Set y-position</td></tr><tr><td>Byte 10</td></tr><tr><td>Byte 11</td></tr><tr><td>Byte 12</td></tr></table></td></tr>
<tr><td>Engine on/off State</td><td>0x3</td><td><u>1 Byte:</u>

0x0: Engine off.
0x1: Engine on.</td></tr>
</table>

5. Nonfunctional Requirements

#001. The logic of Mars rovers should be implemented in SCADE Suite.

#002. Mars rover should be implemented into the TM4C1294NCPDT microcontroller.